

STANDARD OPERATING PROCEDURE			
		PREPARATION OF PGS EMBRYO BIOPSY COLLECTION KITS	
SOP No.:	ADL/SOP/PGS/KIT/001	Effective Date:	
Version No.:	01	Review Date:	

1. PURPOSE:

To describe the standardized preparation, sterilization, buffer aliquoting, assembly, labeling, quality-control and release of PGS embryo-biopsy collection kits intended for receipt and preservation of embryo-biopsy samples prior to downstream whole-genome amplification (WGA) and PGS testing.

2. SCOPE:

This SOP applies to trained and authorized personnel preparing embryo-biopsy collection kits in the designated clean laboratory area. It covers preparation of sterile PCR tubes, DPBS-based preservation buffer, kit packing and lot traceability. It does not describe embryo biopsy, WGA, sequencing or clinical interpretation.

3. ABBREVIATIONS:

ABBREVIATION	FULL FORM
PGS	Preimplantation Genetic Screening
DPBS	Dulbecco's Phosphate Buffered Saline
ICSI	Intracytoplasmic Sperm Injection medium
WGA	Whole Genome Amplification
PVDF	Polyvinylidene fluoride
QC	Quality Control
IFU	Instructions for Use

4. RESPONSIBILITY:

- Laboratory Director / Technical Manager shall approve the procedure and any controlled changes.
- Authorized laboratory personnel shall prepare kits according to the current approved SOP, maintain aseptic technique and complete all records contemporaneously.
- Section supervisor / authorized reviewer shall verify reagent lots, expiry, sterility/QC status, labeling, counts and release of each prepared lot.

5. SAFETY AND CONTAMINATION CONTROL:

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- Wear laboratory coat, gloves and mask as applicable. Change gloves whenever contamination is suspected.
- Perform preparation only in a cleaned designated cabinet/work area. Separate kit preparation from amplified-DNA/post-PCR activities.
- Use DNase/RNase-free, sterile disposables. Avoid touching the inner surface of tubes, caps, racks or bags.
- Use UV only in accordance with the cabinet/equipment safety instructions. Do not expose personnel to UV radiation.
- 70% IPA or 70% ethanol may be used for surface disinfection as per the laboratory-approved cleaning procedure.

6. EQUIPMENT AND MATERIALS:

- Clean cabinet / laminar work area with UV facility
- Calibrated micropipettes and sterile filtered tips
- Vortex mixer
- Refrigerator (2–8°C)
- Sterile 1.5 mL tubes for preservation-buffer aliquots
- Sterile PCR tubes, 0.2 mL, flat cap, DNase/RNase-free (Axygen PCR-02-C or validated equivalent)
- PCR workstation rack with cover – Tarsons Cat. No. 241000 (or validated equivalent)

Product type: PCR Workstation Rack with cover; polypropylene (PP).

Blue rack with transparent lid; cover/lid included.

96 wells (12 × 8); compatible with 0.1 mL and 0.2 mL PCR tubes, strips and plates.

Autoclavable; manufacturer specifies the rack as non-sterile. Clean/decontaminate according to the validated laboratory procedure before kit preparation.

- Sterile zip-lock bags and kit cartons/boxes
- Sterile 15 mL / 50 mL conical tubes as required
- Sterile 0.2 µm PVDF syringe filter, 25 mm (Whatman GD/X Cat. No. 6900-2502 or validated equivalent)
- Gibco DPBS (1X), calcium- and magnesium-free, REF 14190-144 or validated equivalent
- Vitrolife ICSI medium, REF 10111, or the laboratory-validated equivalent

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- Approved labels, kit preparation worksheet, TRF/consent form and embryo-biopsy SOP/instructions

7. REAGENT AND CONSUMABLE CHECKS BEFORE PREPARATION:

- Verify product name, catalogue/reference number, lot number, expiry date and package integrity for all reagents and consumables.
- Do not use expired, damaged, visibly contaminated or improperly stored material.
- Record DPBS lot, ICSI-medium lot, PCR-tube lot, Tarsons PCR-rack lot/batch (where available), and filter lot in the preparation record.
- The supplied material photographs/product information show DPBS REF 14190-144, Whatman GD/X 0.2 µm PVDF filter Cat. No. 6900-2502, Axygen PCR-02-C PCR tubes, Vitrolife ICSI REF 10111, and Tarsons PCR Workstation Rack with cover Cat. No. 241000. Use the current approved/validated materials at the time of preparation.

8. PROCEDURE:

8.1 Work-area preparation

1. Remove unnecessary materials from the cabinet/work area.
2. Clean the work surface with 70% IPA or 70% ethanol according to the approved laboratory cleaning procedure.
3. Wipe the external surfaces of racks, outer cartons/boxes, zip-lock bags and other materials before placing them in the clean area.
4. After cleaning and with the cabinet empty, expose the work area to UV for 20 minutes (or the validated cabinet cycle). Switch UV off before personnel begin work.

8.2 Arrangement and UV treatment of kit components

5. Arrange PCR-tube racks/boxes inside the clean cabinet.
6. Place fifteen (15) sterile 0.2 mL PCR tubes in each designated rack/kit box, unless a different count is specified in the current approved kit configuration.
7. Arrange required sterile conical tubes, racks and other UV-compatible components with caps/openings positioned as specified by the validated local procedure.
8. Expose arranged UV-compatible materials to UV for 40 minutes (or the validated cycle). Do not UV-irradiate reagents unless specifically validated.

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9. After UV treatment, switch UV off and proceed using aseptic technique.

8.3 Preparation of preservation buffer

10. Transfer 20 mL DPBS (1X) into a sterile 50 mL conical tube.
11. Add 25 µL of ICSI medium to the DPBS and mix by gentle vortexing. NOTE: this volume is transcribed from the supplied preparation note and must be verified against the laboratory-approved validation/worksheet before controlled release of this SOP.
12. Keep the prepared buffer at 2–8°C during preparation when not actively handling it.
13. Immediately before filtration, use sterile filtered tips and a sterile syringe/filter assembly.
14. Filter the prepared preservation buffer through a sterile 0.2 µm PVDF filter into a sterile receiving tube using aseptic technique. Do not reuse the filter.
15. Label the filtered buffer with preparation date, preparer initials, component lot numbers and assigned preparation-lot number.

8.4 Aliquoting of preservation buffer

16. Label sterile 1.5 mL tubes according to the kit/lot plan.
17. Aliquot 800 µL of sterile-filtered preservation buffer into each 1.5 mL tube.
18. Close each tube immediately after aliquoting.
19. Visually inspect for leakage, particulate matter, turbidity or abnormal appearance. Reject any affected aliquot.
20. Keep aliquoted preservation buffer at 2–8°C until kit assembly/release, unless the validated stability study specifies another condition.

8.5 Kit assembly and packing

21. For each kit box, place the required rack containing fifteen (15) sterile 0.2 mL PCR tubes together with two labeled 1.5 mL preservation-buffer tube (800 µL), unless otherwise defined by the approved kit configuration.
22. Place each assembled kit in the designated clean protective packaging.
23. Pack five (5) parafilm pieces into small zip-lock bag, as described in the supplied preparation note.
24. Assign a kit lot number. For a batch of 10 kits, sequential numbering may be used (example: 1–10 followed by the preparation date/month-year code) according to the laboratory lot-numbering convention.

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25. Include the applicable TRF/consent form, Form G (where required) and embryo-biopsy collection SOP/instructions with the dispatch pack.

26. Seal the final package and record the total number of kits prepared and released.

9. QUALITY CONTROL / STERILITY CHECK:

- From each newly prepared preservation-buffer lot, retain a representative aliquot for QC as defined by the laboratory quality plan.
- Submit an aliquot for WGA preparation/microbiological sterility/growth assessment for bacterial and fungal contamination, as stated in the supplied preparation note.
- Do not release the preparation lot for clinical sample collection until required QC/sterility acceptance is documented, unless an approved risk-based release procedure exists.
- If growth/contamination is detected, quarantine the affected lot, investigate the source, document nonconformity and repeat preparation with new sterile materials/reagents as appropriate.

10. ACCEPTANCE / REJECTION CRITERIA:

PARAMETER	ACCEPTANCE	REJECTION / ACTION
Materials	Correct product/lot, within expiry, intact packaging	Expired, damaged, incorrect or untraceable material
Buffer appearance	Clear; no visible particulate/turbidity	Turbidity, precipitate, particulate matter or leakage
Aliquot volume	800 µL per 1.5 mL tube	Incorrect volume outside validated tolerance
PCR tubes	15 sterile 0.2 mL tubes per kit/rack	Wrong count, damaged/open/contaminated tubes
Sterility QC	No bacterial/fungal growth according to approved microbiology method	Any growth: quarantine/reject lot and investigate

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Documentation	Complete lot, date, preparer/reviewer and component traceability	Incomplete or non-traceable records
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11. STORAGE AND DISPATCH:

- Store prepared preservation-buffer aliquots/kits at the laboratory-validated temperature. The supplied preparation note indicates 4–8°C during preparation/storage of buffer.
- Protect kits from contamination, leakage, direct sunlight and temperature excursions.
- Define kit expiry only from an approved stability/validation study; do not assign an arbitrary expiry solely from component expiry dates.
- Record dispatch date, kit lot, quantity and receiving client/location for full traceability.

12. RECORDS:

- PGS Kit Preparation Batch Record
- Reagent and consumable lot/expiry record
- Cleaning and UV record
- Preservation-buffer preparation and aliquoting record
- Microbiology sterility/QC report
- Kit release and dispatch log
- Deviation/nonconformity record, if applicable

13. REFERENCES:

- Current manufacturer IFU/product information for Gibco DPBS (1X), REF 14190-144.
- Current manufacturer IFU/product information for Vitrolife ICSI medium, REF 10111.
- Current manufacturer product information for Whatman GD/X sterile 0.2 µm PVDF filter media, Cat. No. 6900-2502.
- Current manufacturer product information for Axygen 0.2 mL PCR tubes, PCR-02-C.
- Current manufacturer product information for Tarsons PCR Workstation Rack with cover, Cat. No. 241000.
- Applicable laboratory biosafety, cleaning, environmental monitoring, sterility-testing, PGT sample-receipt and quality-management procedures.

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ANNEXURE 1 – PGS KIT PREPARATION BATCH RECORD

Preparation Date		Kit Lot No.	
No. of Kits Prepared		No. Released	
DPBS Lot / Expiry		ICSI Lot / Expiry	
PCR Tube Lot / Expiry		Tarsons Rack Lot / Batch	
Buffer Volume Prepared		Aliquot Volume	800 µL
UV Cycle – Initial		UV Cycle – Components	
Sterility QC Result		QC Report No.	
Prepared By / Sign		Reviewed By / Sign	

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